

Roll No.

TME-302

B. TECH. (ME) (THIRD SEMESTER)
MID SEMESTER EXAMINATION, Oct., 2022

MATERIAL SCIENCE

Time : 1½ Hours

Maximum Marks : 50

Note : (i) Answer all the questions by choosing any *one* of the sub-questions.

(ii) Each sub-question carries 10 marks.

1. (a) Draw the figures and explain the stacking in C.C.P. and H.C.P. structures. (CO1)

OR

- (b) Explain briefly the terms "Atomic packing factor and diffusion". (CO1)

2. (a) Explain briefly the different types of point defects with diagrams. (CO4)

OR

- (b) Derive an expression for the relation between the atomic radius and lattice constant for F.C.C. and B.C.C. structures. (CO4)

3. (a) What is a line dislocation defect ? Is it desirable to have this defect in crystal structures ? (CO1)

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(2)

OR

(b) Explain the mechanism of creep with the neat and labelled diagrams.

(CO1)

4. (a) Differentiate between the elastic deformation and plastic deformation with the help of stress strain curve for mild steel.

(CO4)

OR

(b) Define the terms "Strength", "Brittleness", "Ductility", "Hardness" and "Malleability".

(CO4)

5. (a) Draw and explain the "S-N" curve for fatigue.

(CO4)

OR

(b) Differentiate between the fatigue and creep on the basis of their behavior in materials.

(CO4)